A CASE-STUDY BASED REPORT

ON

Plant Leaf Disease Detection Using Machine Learning

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MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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PLANT LEAF DISEASE DETECTION USING MACHINE LEARNING

A case-study based report on the application of deep learning to classify plant leaf images into healthy and diseased categories supported by the selected dataset.

Abstract

Plant diseases can affect crop quality and productivity, and identifying symptoms early can help growers decide when to seek further advice. This project develops a plant leaf image-classification system using MobileNetV2 transfer learning with TensorFlow/Keras and the PlantVillage dataset. The dataset contains 38 supported classes. The trained model was evaluated on 10,861 test images and achieved 95.35% test accuracy. A Streamlit web application allows a user to upload a leaf image and view the predicted class and confidence scores. The system is intended for educational and preliminary screening use; predictions should not replace expert assessment.

1. Introduction

1.1 Background

Plant diseases may reduce crop quality and productivity. Leaf symptoms such as spots, discoloration, or changes in texture can provide useful clues, but interpreting them manually may require experience. Image classification with machine learning offers a way to analyze visual patterns and suggest a likely class from examples seen during training.

1.2 Motivation for the Project

The project aims to make preliminary plant-leaf classification faster and easier to access. A simple image-upload interface can help users explore a model prediction without needing to run training code or understand the model implementation.

1.3 Overview of the Work Carried Out

The work covers preparing the PlantVillage image dataset, training a transfer-learning model, evaluating its performance, saving the model and class labels, and building a Streamlit interface for image prediction. Evaluation outputs include training curves, a confusion matrix, a classification report, and a training log.

1.4 Scope and Responsible-Use Statement

The system classifies images into 38 classes represented in the selected dataset. It performs best when an input image resembles the training data and shows a clear leaf. Images from unfamiliar crops, diseases, backgrounds, lighting conditions, or camera types may produce unreliable predictions. The output is a model estimate, not a confirmed diagnosis or a substitute for advice from an agricultural expert.

2. Problem Statement

2.1 Case-Study Scenario

A grower or student observes a possible problem on a plant leaf and wants an initial indication of which supported category the image resembles. The project explores whether an image-classification model can provide a useful preliminary prediction through a straightforward web interface.

2.2 Central Case-Study Question

Can a MobileNetV2-based transfer-learning model classify plant leaf images into the supported PlantVillage classes with useful test performance, and can its predictions be made accessible through a web application?

2.3 Formal Problem Definition

This is a supervised multi-class image-classification task. Given a leaf image x, the model produces a predicted class ■ from the set of 38 supported classes. The prediction is selected from the model's learned class scores. The system is evaluated against labelled test images using accuracy, precision, recall, and F1-score.

2.4 Objectives that Guide the Study

- Build a plant leaf image classifier using transfer learning.
- Use TensorFlow/Keras and MobileNetV2 to learn visual features.
- Evaluate the trained model on labelled test images using standard classification metrics.
- Provide a Streamlit interface for image upload and prediction display.
- Record evaluation artifacts to support inspection of model behaviour.

2.5 Assumptions and Limitations

- The uploaded image is readable and depicts a plant leaf relevant to one of the model's supported classes.
- The class labels in the dataset are used as the reference categories during training and evaluation.
- Test-set performance does not guarantee the same performance on photographs taken in real fields.
- The system may be affected by image quality, lighting, background, overlapping symptoms, and classes not represented in training.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project uses the PlantVillage dataset, which contains labelled images of healthy and diseased plant leaves. The implemented classifier supports 38 classes. The local dataset occupies approximately 1.3 GB. During evaluation, the test data loader reported 10,861 images across 38 classes.

3.2 Purpose of EDA

Exploratory data analysis (EDA) helps understand the structure of the dataset before interpreting model results. It can be used to inspect class labels, image availability, class distribution, sample image quality, and possible challenges such as visually similar categories.

3.3 Preparation Steps Actually Performed

- Prepared the PlantVillage dataset locally for use by the training and evaluation scripts.
- Loaded labelled images through the data pipeline and used the class labels for classification.
- Used the model's image-preprocessing pipeline for training and prediction.
- Saved the class names in `models/class_names.json` and the trained model in `models/plant_disease_model.keras`.
- Kept evaluation artifacts in the `results` folder.

3.4 EDA Exploration – Implementation

The project's workflow includes checking dataset availability and class names, loading images, inspecting training and evaluation outputs, and reviewing the generated confusion matrix and training curves. The evaluation report provides class-wise precision, recall, and F1-score. A detailed class-count table is not asserted here because no verified per-class counts were supplied for this report.

3.5 Summary of EDA Findings

The dataset loader identified 38 classes and 10,861 test images. The project generated training curves, a confusion matrix, an evaluation report, and a CSV training log. These outputs help review overall performance and class-level errors. Further testing on field photographs is necessary to understand how well the system generalizes beyond the dataset.

4. Algorithm Selection and Implementation

4.1 Algorithm Selection

MobileNetV2 was selected as the base model for transfer learning. It provides reusable image features while being relatively lightweight compared with many larger image-classification architectures. Transfer learning allows the project to adapt learned visual representations to the plant-leaf classes.

4.2 Implementation

The model workflow uses Python and TensorFlow/Keras. The training script prepares the data pipeline and trains the classifier; the evaluation script calculates metrics and generates reports; the prediction script classifies an individual image; and the Streamlit application presents image upload and prediction results. The saved model and class-name file are used by the prediction workflow.

4.3 Evaluation Results

The evaluated model achieved 95.35% test accuracy on 10,861 test images. The reported macro-average precision was 0.95, recall 0.94, and F1-score 0.94. The weighted-average precision was 0.96, recall 0.95, and F1-score 0.95. These results describe performance on the evaluated test set and should not be interpreted as a guarantee for every real-world image.

4.4 Application Output

The Streamlit interface lets a user upload a leaf image and view the model's predicted class with confidence scores for leading predictions. Confidence scores express the model's relative output, not certainty that a disease is present. Users should seek qualified agricultural guidance before making crop-treatment decisions.

5. Conclusion

This project demonstrates a deep-learning workflow for plant leaf disease classification using MobileNetV2 transfer learning, TensorFlow/Keras, the PlantVillage dataset, and a Streamlit interface. The model achieved 95.35% test accuracy across 10,861 evaluated images from 38 classes. Training curves, a confusion matrix, and a classification report provide supporting evaluation outputs. The project meets its goal of making model predictions accessible through a simple interface, while its reliability on real-world field images remains an important area for further testing and improvement.

Note: The dataset and test results describe the project's current implementation. The model should be used for learning and preliminary screening, not as the sole basis for agricultural decisions.